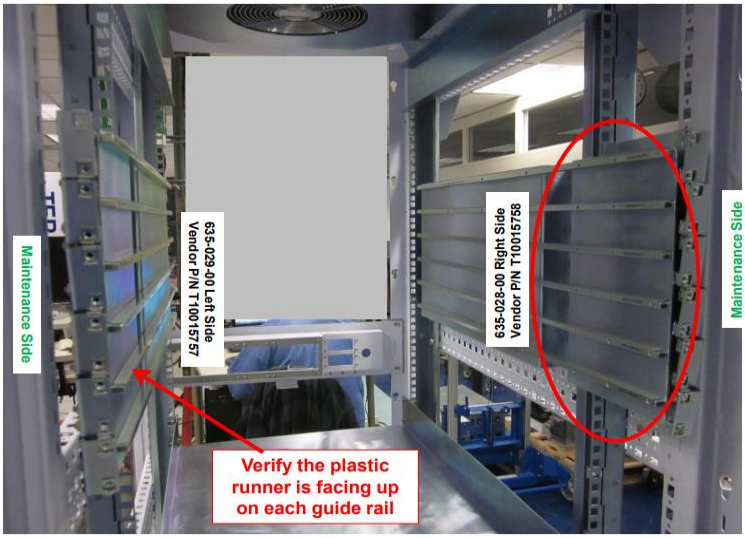
Build Overview:

1 **Cabinet Prep**
2 **SMC rails Install**
3 **ADB Shelf Install**
4 **Fan Install**
5 **PSU brackets Install**



CHANGE HISTORY

Date	BPM Rev	Change Details	Updated by
	1	New Issue	JOSEPH AMAD

Prepared By:
JOSEPH R. AMAD
Manufacturing Engineer

Checked By:
IVAN NOWELL PANIQUE
Team Lead

Approved By:
SHERWIN ARIOSTE
Operations Manager

METHOD:

Flow chart of operation steps

BPM preparation	See (1. Scope of application)
Preparation Tools	See (2. Tools/Materials/Parameters---1. Tools)
Prepare Materials	See (2. Tools/Materials/Parameters---2. Materials)
Complete Operation	See (3. Operation Steps)
Inventory	See (2. Tools/Materials/Parameters---2. Materials)
Operation Check	See (4. Matters needing attention)
Inventory collection tools	See (2. Tools/Materials/Parameters---1. Tools)
Inventory collections tools	See (2. Tools/Materials/Parameters---1. Tools)

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM NO.	PART NUMBER	PART NAME
1	661-717-00	System Label
2	479-133-42	China Rohs Label
3	406-512-12	M5 Screws
4	672-871-00	Emo Bracket
5	813-929-01	Short Velcro
6	628-453-00	UF Plus 24 Slot Support Cabinet Chasis
7	632-461-00	CDU Sliding Plates
8	470-483-04	SCRW CP BUT HD M5X
9	360-812-01	NUT, CAGE M5, ZN PLATE
10	574-777-00	SMC RAILS
11	472-426-12	M5 SCREW
12	635-920-00	CABINET FAN TRAY
13	603-096-00	Nut Hex M8
14	637-537-00	Movable Brackets
15	637-538-00	Fixed Brackets
16	635-028-00	PSU Brackets (right)
17	635-029-00	PSU Brackets (left)

MACHINE:

1. Manual Torque 1.6Nm, 3.3Nm,
2. Torque Wrench 3.3Nm
3. Flat Screwdriver
4. Engineering Stair
5. Bit PH2, Socket 8mm
6. Wrench 19mm

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	628-453-00	UF PLUS 24 Slot Support Cabinet Chasis	1

MACHINE: N/A

METHOD: Support Cabinet Preparation

2. Check the Cabinet Chasis (628-453-00). Make sure the cabinet clear from coating defects, scratches, bent, deform, EMI gasket not torn and etc.

3. There was a key to unlock the cabinet skin. To unlock, the key must turn clockwise and to lock the key turn counter clockwise. After that, hold at the slot and lift up the skin to take out.

4. Save the skin that have done take out and send it to button up station

After take out cabinet skin

Before take out cabinet skin

Example of Reject :

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: **N/A**

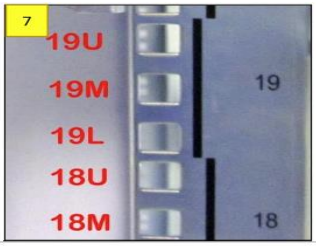
MACHINE: **N/A**

METHOD:

5. The operation side of the cabinet as shown in the figure-the operating direction of the cabinet panel is also defined as the front of the cabinet.
6. The cabinet maintenance side as shown in the figure-the direction of the cabinet facilities on the bottom panel is also defined as the back of the cabinet.
7. The position of the guide rail is defined by numbers, which are printed on the guide rail. The positions of the numbers are U (upper), M (middle), and L (lower).

Operation side

Maintenance side



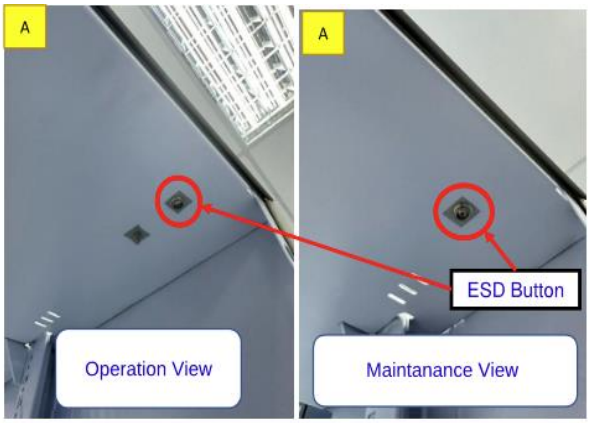
Position Guide Rail

MAN: :CERTIFIED OPERATOR, TECHNICIAN

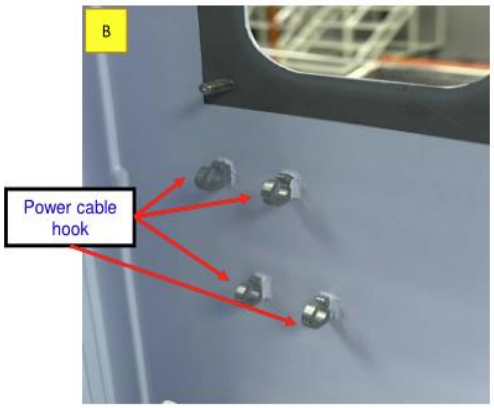
MATERIAL: **N/A**

MACHINE: **N/A**

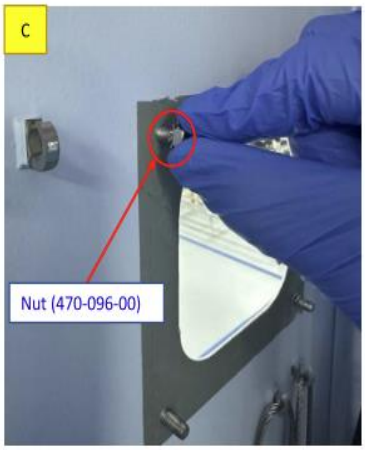
METHOD: Inspection Inside the Cabinet



A. Make sure surface of the ESD button doesn't have any foreign material or coating and button fully tighten



B. Check the power cable hook installed in good condition and not reversed.



C. Check all stud thread at the 4 hole inside the cabinet using 1pcs nut (470-096-00) and make sure no thread stuck when checking.

D. Check all stud thread at cabinet chassis door skin holes with 1pcs nut (470-096-02). There were total 20 places to check. 4 at Maintanance side, 8 for left and right side and 8 for Operation side. Make sure all thread are smooth and clean

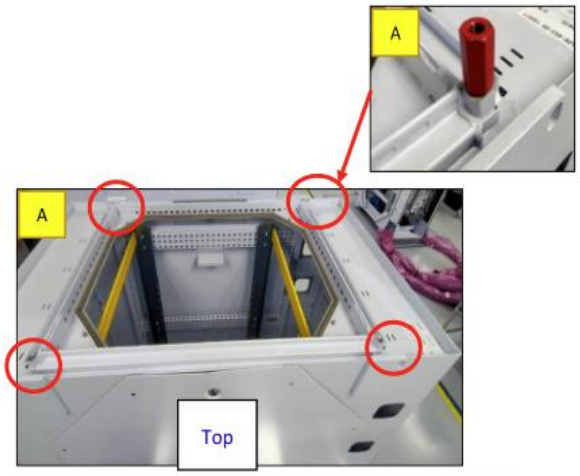
MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

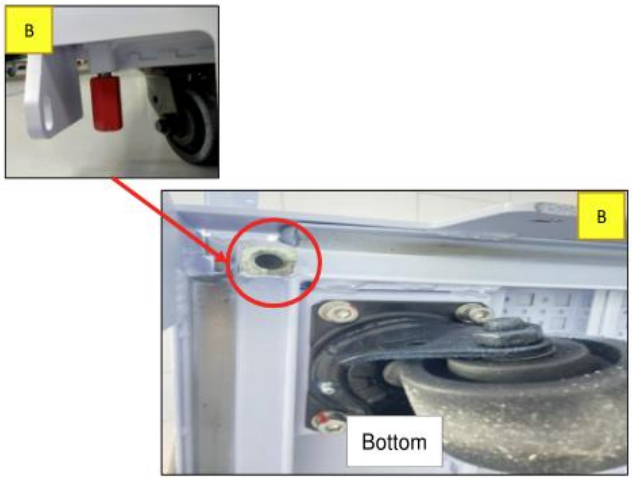
MACHINE: N/A

METHOD:

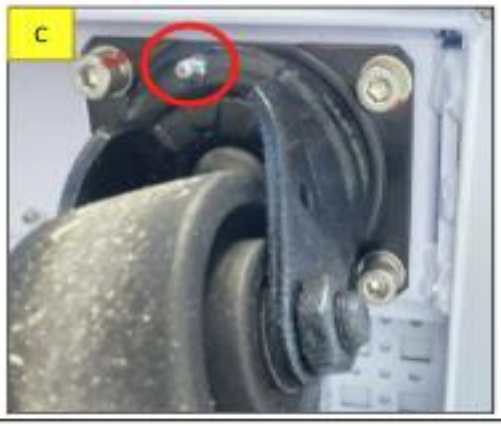
9. Take 1 Cabinet Leg Engineering Tool and install it each corner of the cabinet to check the screw thread. (Top and Bottom)



A. Check the thread at the top of cabinet chasis. There were four places as shown in diagram. Make sure the thread are smoothly and no rust.



B. Check the thread at the bottom of cabinet chasis. There were four places as shown in diagram. Make sure the thread are smoothly and no rust.



C. Make sure that all four wheels have a pinas shown in diagram.

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

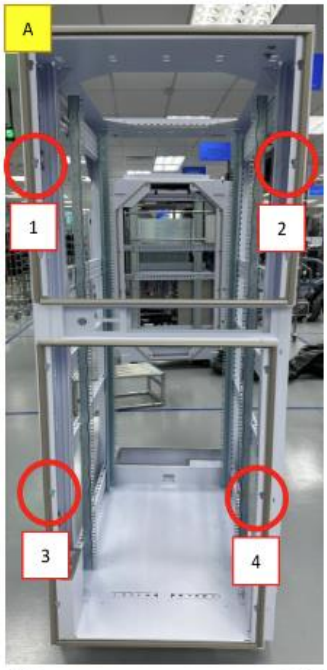
ITEM	P/N	DESCRIPTION	QTY
1	674-565-01	Door Ground Connection	10
2	603-096-02	Nut 7mm	20

MACHINE: Torque Wrench 1.6Nm, Socket 7mm

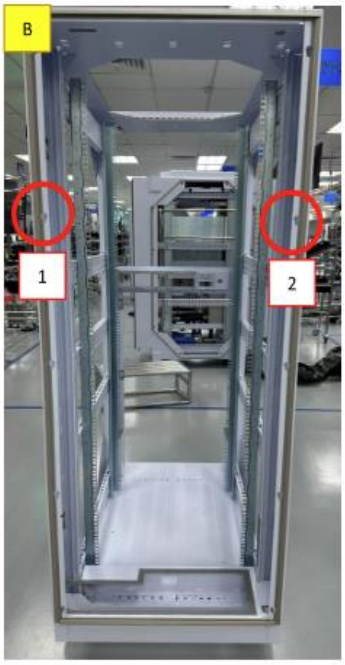
METHOD: Installation of the door ground connection

1. Install door ground connection (674-565-01) qty 10 using nuts (603-096-02) - Torque 1.6N.m.

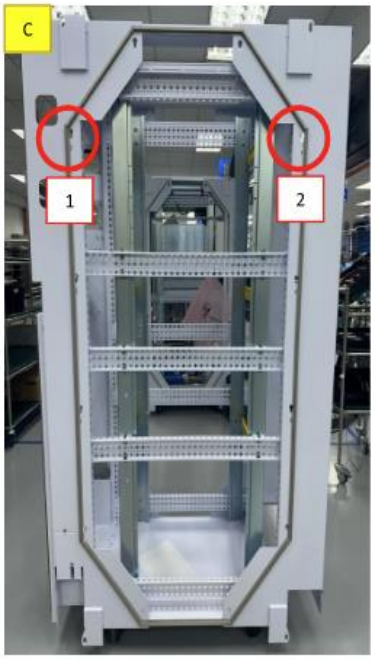
Example shown orientation of Door Ground (674-565-01)



A. From **Operation Side**, Install **4pcs** Door Ground (674-565-01) at circle shown



B. From **Maintanance Side**, Install **2pcs** Door Ground (674-565-01) at circle shown.



C. From **Right Side** of cabinet, Install **2pcs** Door Ground (674-565-01) at circle shown.

D. From **Left Side** of cabinet, Install **2pcs** Door Ground (674-565-01) at circle shown.

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	632-461-00	CDU Sliding Plates	2
2	470-483-04	SCRW CP BUT HD M5X	8

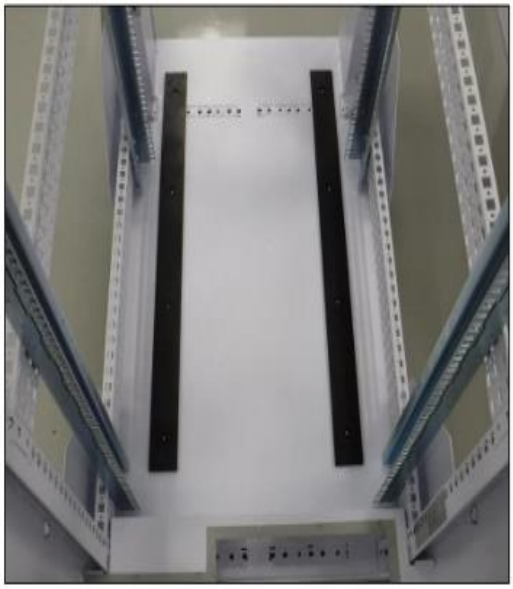
MACHINE: Torque 3.3Nm, Bit Hex 3mm

METHOD: Installing CDU Sliding Plate

1. Install 2pcs CDU sliding plates (632-461-00) at the bottom of the cabinet, and fix them with 8pcs M5 screws (470-483-04).
2. The torque value is 3.3Nm and 3mm hexagon bit.

Before installation, check M5 screw thread can go through thread holes smoothly. The figure shows from maintanance surface view.

CDU sliding plate (653-631-00) and M5 screws(470-483-04)



Position of the CDU sliding plate when install at cabinet

Position of the screw M5

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	360-812-01	NUT, CAGE M5, ZN PLATE	39

MACHINE: Flat Screwdriver

METHOD: Installing Nut Cage to Cabinet Rail

1. Install 20 nuts (360-812-01) on the guide rails on both sides of the cabinet maintenance surface
2. Install 18 nuts (360-812-01) on the guide rails on both sides of the operation surface of the cabin
3. Operation Surface of cabinet, Install 1 nuts (360-812-01) on the left side of the guide rail.

Maintenance	Operation
20M, 21M,22M, 23M, 24L,25U, 28U, 33L, 37U,38L	3U, 17U, 22M, 23M,36L,36U, 37M,37U, 38L
	18 pcs
	28M (only left guide rail)
20 pcs	1 pcs

Postion of the nut cage (360-812-01)

Operation surface - Total nut 19

Maintanance surface - Total nut 20

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	574-777-00	SMC RAILS	2
2	472-426-12	M5 SCREWS	8

MACHINE: Torque 3.3Nm, Bit PH2

METHOD: SMC Rail Installation

1. Install 2pcs SMC Rails (574-777-00) in location 37U and 38L on both sides of cabinet using 4pcs M5 Screws (472-426-12) each smc rail – Torque to 3.3Nm and PH2 Bit.

NOTE: Square Holes in the rails should be oriented to the **Maintenance** side of the cabinet

A. Use 3.3Nm torque with PH2 Bit to securethe SMC Rail.

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	574-777-00	Door Ground Connection	10
2	472-426-12	Nut 7mm	20

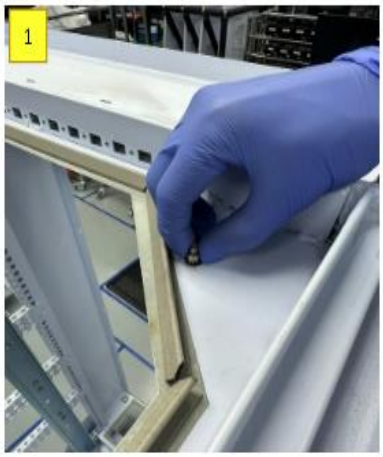
MACHINE: Torque 3.3Nm, Bit PH2

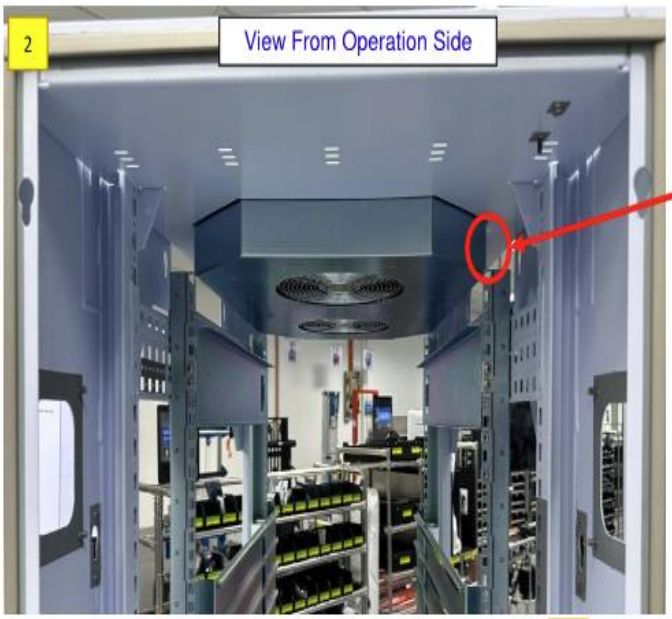
METHOD: Installation of the Cabinet Fan Tray

1.Remove the nuts and flat washer that come with the cabinet and discard the item.

2. Install 1pcs Fan Tray Assembly (635-920-00) in place of the top cover with the fans facing out and the cable hole oriented on the Operator right side of the cabinet.

3.Using M5 Nut Hex 4 pcs (603-096-00) to hold the fan tray – Torque to 3.3Nm and 8mm Sokcet





Cable Hole

3. Use 3.3Nm torque and 8mm Socket to secure 4pcs nut.

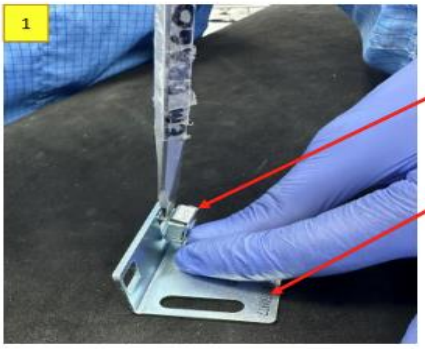
MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	360-812-01	M5 Cage Nuts	24
2	637-537-00	Moveable Brackets	12
3	637-538-00	Fixed Brackets	2
4	603-096-00	Nut Hex M8	24

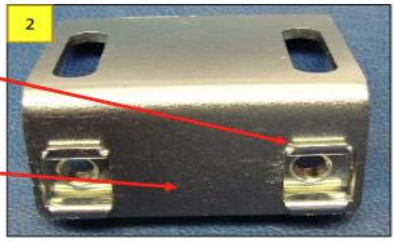
MACHINE: Flat Blade Screwdriver

METHOD: Installing PSU Brackets



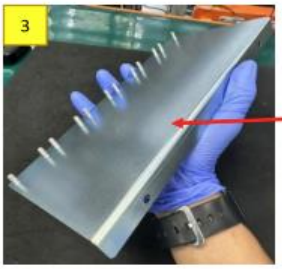
Cage Nut (360-812-01)

Moveable Bracket (637-537-00)



2. Diagram shown the orientation of the Nut Cage (360-812-01)

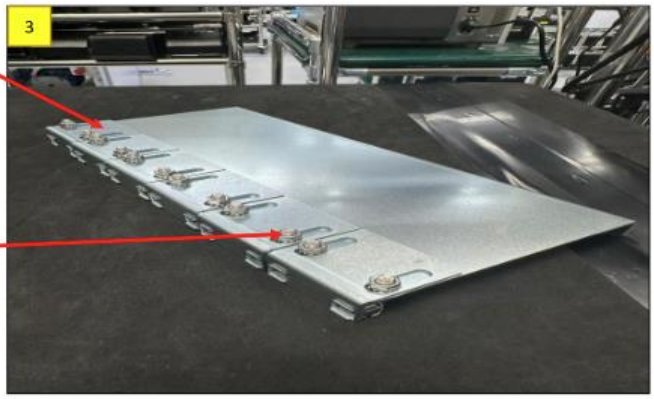
1.Install 2pcs M5 Cage Nuts (360-812-01) on 1pcs Movable Brackets (637-537-00). Diagram shown how to install.



Moveable Bracket (637-537-00)

Fixed Bracket (637-538-00)

M5 Nut (603-096-00)



3. Install Moveable Brackets (637-537-00) that installed before onto Fixed Brackets (637-538-00), 6pcs (637-537-00) per bracket (637-538-00), using 12pcs M5 lock nuts (603-096-00) per bracket (637-538-00) – Do not torque,Finger tight only.

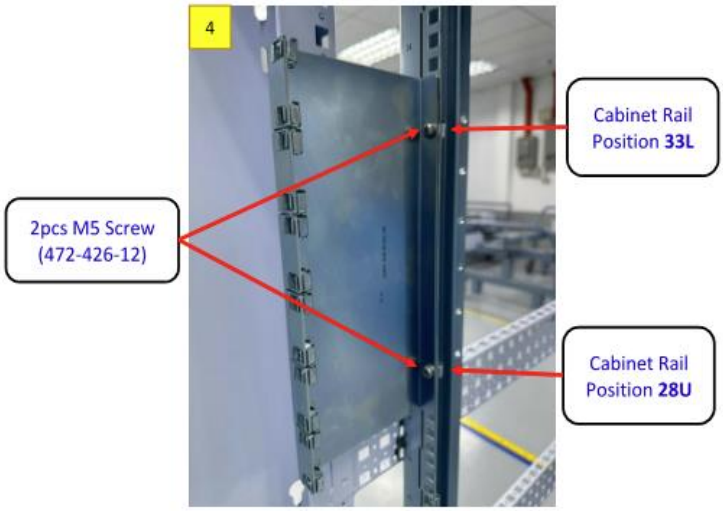
MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	637-538-00	Fixed Brackets	2
2	472-426-12	M5 Sems Screws	4

MACHINE: Torque 3.3Nm, Bit PH2

METHOD: Install PSU Brackets



4. At **maintanance side**, install the Fixed Bracket (637-538-00) that assemblebefore to cabinet rail at postion **28U** and **33L** for both side of cabinet rail using 2pcs M5 Screw (472-426-12).

A. Torque 2pcs M5 Screw (472-426-12) to 3.3Nm with PH2 Bit. Torque to both side rail.

B. Diagram shown after installed the PSU Brackets.

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	635-028-00	PSU Brackets (right)	1
2	635-029-00	PSU Brackets (left)	1

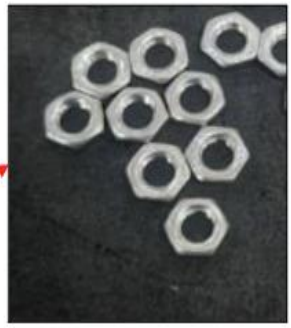
MACHINE: 8mm Socket

METHOD: Install PSU Guide Rail

1.Obtain the PSU Guide Rail (right) 635-028-00 and PSU Guide Rail (left) 635-029-00

2.Remove and discard 8pcs incoming nuts at each PSU Guide Rail

Note : Can use 8mm socket if M5 nut too tight when to unscrew it



This nut will be discard.

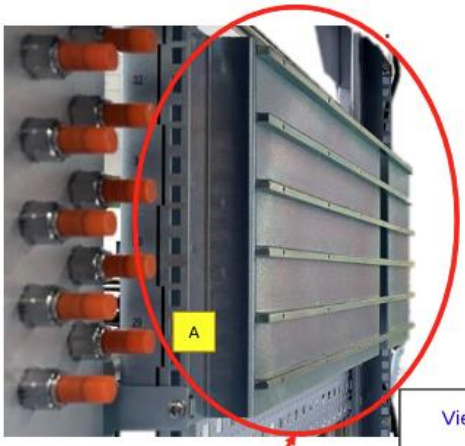
MAN: CERTIFIED OPERATOR, TECHNICIAN

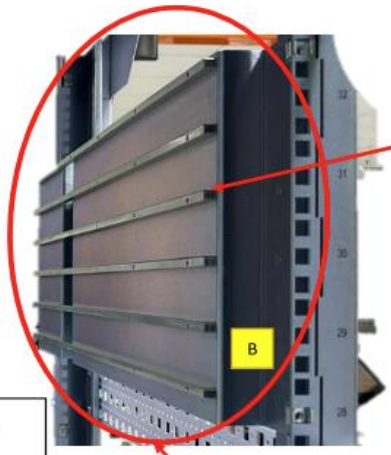
MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	603-096-00	M5 Nuts	16

MACHINE: Torque 3.3Nm, 8mm Socket

METHOD: Install PSU Guide Rail



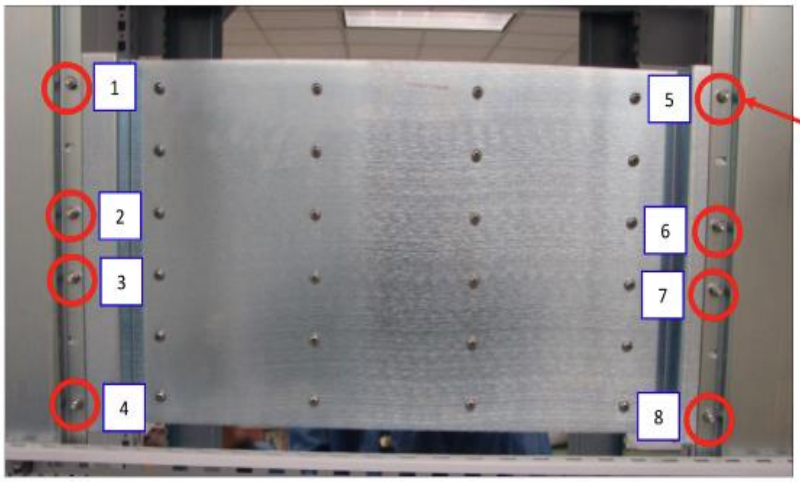


Confirm that the plastic chanel direction is **upward** at the guide rail.

View From **Operation Cabinet**

A. Install the PSU bracket (635-028-00) on **the left side of the cabinet**, and fix it with 8 M5 nuts (603-096-00). The tool is an 8mm socket wrench, The torque value is 3.3 N·m.

B. Install the PSU bracket (635-029-00) on the **right side of the cabinet**, and fix it with 8 M5 nuts (603-096-00). The tool is an 8mm socket wrench, The torque value is 3.3 N·m.



Fix the PSU Guide Rail with 8pcs nut each rail by using socket 8mm, 3.3Nm torque. Total 16pcs both